TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: CP_SEP490 **Duration time:** From 05/01/2026 To 30/04/2026

Profession: Information Technology **Specialty:** Software Engineering

Kinds of person make registers: Lecturer: ☑ Students: ☐

1. Register information for supervisor

	Full name	Phone	E-Mail	Title
Supervisor 1	Nguyễn Thị Hạnh	0935688515	Hanhnt54@fe.edu.vn	MSc
Supervisor 2				

2. Register information for students

	Full name	Student code	Phone	E-mail	Role in Group
Student 1					Leader
Student 2					Member
Student 3					Member
Student 4					Member
Student 5					Member

3. Register content of Capstone Project

3.1. Capstone Project name:

English**:** Develop **BusDN** – a real-time bus management and tracking system in Da Nang City using React, Nodejs, Mongodb .

Vietnamese**:** Xây dựng BusDN – Hệ thống quản lý và theo dõi xe buýt theo thời gian thực tại Thành phố Đà Nẵng sử dụng React, Nodejs, Mongodb *(SV có thể lựa chọn công nghệ khác).*

Abbreviation**:** PCLC

3.2. Context: (brief introduction)

***Objectives**:*

In the context of urban digital transformation and smart city development, the demand for intelligent public transportation management has significantly increased. Currently, passengers in Da Nang City face difficulties in accessing real-time information about bus routes, schedules, and arrival times, while transportation operators need an effective system for fleet management and performance monitoring.

BusDN aims to provide a unified digital platform that enables real-time tracking, management, and optimization of bus operations in Da Nang City. The system allows passengers to conveniently track bus locations and arrival times, while administrators and drivers can monitor operations, manage routes, and enhance service reliability.

Technology/algorithm:

- Front-end: React.js / Next.js/…
- Back-end: ASP.NET Core / Java Spring Boot, Node.js . …
- Database: Firebase Realtime Database, MongoDB Atlas.,…
- Version Control: GitHub
- CI/CD Tools:
  - GitHub Actions / GitLab CI/CD (for automated testing and deployment)
  - Docker / Kubernetes (for containerization and scalable deployment)
  - AWS / Firebase / Vercel (for cloud hosting and deployment)

3.3. Summarize the contents to be researched and the expected outputs of the project:

- **System and User Analysis**

- Analyze requirements from three user groups: passengers, drivers, and administrators.

- Study the current public transportation challenges in Da Nang.

- **Technology Research**

- Research GPS tracking and real-time data synchronization methods.

- Evaluate the use of Firebase and Google Maps API for real-time updates.

- **System Design and Architecture**

- Design the overall architecture for client–server communication.

- Develop data models for vehicles, users, and routes.

- Build wireframes and UX flows for the mobile and web applications.

- **Implementation and Integration**

- Develop a mobile application for passengers and drivers.

- Implement an admin web dashboard for monitoring and management.

- Integrate GPS data for live tracking and route visualization.

- **Testing and Evaluation**

- Test the accuracy of real-time tracking and route updates.

- Evaluate system usability, reliability, and scalability.

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- Website application: ☑
- Mobile application: ☑
- iOT: ☐
- Game: ☐
- Research: ☐
- N/A: ☐

3.4. Expected features

Passenger Functions

User Registration and Login

- Register using email, phone number, or social media accounts.
- Verify identity via OTP for security.

Search for Bus Routes

- Search by bus number, station name, or destination.
- Display route details, operating hours, and frequency.

Real-time Bus Tracking

- Display current bus locations on a map (via Google Maps API).
- Show estimated arrival times (ETA) for each stop.
- Send notifications when the selected bus is approaching the stop.

E-Ticket Booking and Payment

- Choose route, departure, and destination stops.
- Pay using e-wallets, bank cards, or QR code scanning.
- Receive an **electronic ticket (E-Ticket)** containing a QR code or transaction ID.

Ticket and Transaction Management

- View active, expired, or completed tickets.
- Access payment history and transaction details.
- Request refund or ticket change if necessary.

Ratings and Feedback

- Rate bus routes or driver performance.
- Send feedback or report issues directly through the app.

Notifications and Alerts

- Receive alerts about promotions, route changes, or delays.
- Reminders for expiring tickets or top-up suggestions.

Driver Functions

Driver Login

- Log in using credentials provided by the transport company.

Trip Management

- View assigned routes, shift schedules, and trip information.
- Update trip status (in progress, paused, completed).

GPS Location Update

- Automatically send live GPS coordinates to the server at regular intervals.

QR Code Ticket Scanning

- Use the device camera to scan passenger e-ticket QR codes.
- Verify ticket validity and record boarding passengers.

Notifications from Control Center

- Receive messages, route change notifications, or maintenance alerts.

Administrator / Operation Center Functions

User and Driver Management

- Add, edit, or delete passenger, driver, and admin accounts.
- Assign roles and access permissions.

Bus Route and Stop Management

- Create, modify, or delete routes and bus stops.
- Define GPS paths and assign vehicles to routes.

Real-time Fleet Monitoring

- View all buses on a real-time map.
- Monitor operational status (active, idle, delayed, maintenance).

Revenue and Ticket Management

- Generate reports on sold tickets by route, day, or month.
- Track revenue and payment methods.
- Monitor the ratio of e-ticket usage versus manual ticketing.

Data Reports and Analytics

- Export reports (PDF/Excel) on revenue, passenger volume, and delays.
- Filter statistics by route, driver, or time period.

User Feedback Management

- View, respond to, and manage passenger feedback or complaints.
- Record issues and propose corrective actions.

Promotion and Fare Policy Management

- Create discount codes or promotional campaigns.
- Update and manage dynamic fare tables.

Future Enhancement Features

AI-based ETA Prediction – Improve arrival time accuracy using machine learning.

Smart Route Recommendation – Suggest optimal routes based on user travel patterns.

Da Nang, 23 /11/2025

Supervisor (If have)	**On behalf of Registers**
(Sign and full name)	*(Sign and full name)*
Nguyễn Thị Hạnh	